



RAMCO INSTITUTE OF TECHNOLOGY

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NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business Systems

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. CSBE Course Code & Title:

AD3491 Fundamentals of Data Science and Analytics

Name of the Faculty member (s): Mrs.M.Jeya Sundari, AP/CSBE

Innovative Practice Description

- **Unit / Topic: Unit I / Build the Model**

- **Course Outcome: CO1**

- **Topic Learning Outcome: TLO1**

- **Activity Chosen: Concept Map**

- **Justification:**

It is inherently interconnected with data collection and preprocessing, and it is subsequently evaluated using specific metrics. This entire process emphasizes a systematic approach to deriving insights and making data-driven decisions. Hence a concept map activity is given to them to provide a way to summarize all the concepts involved in Data Analytics.

- **Time Allotted for the Activity: 10 Minutes**

- **Details of the Implementation:**

- The activity was conducted at the last 10 minutes of the class after discussing the various Analytics concepts.
- The students are asked to prepare a concept map by depicting the different concepts involved in Data Science.
- The students recollect the concepts involved in FDSA and present the concepts in a easily understandable way.
- Fig.1 shows the concept maps drawn by the students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	3	2	2	1	2	1	2	3	1	1

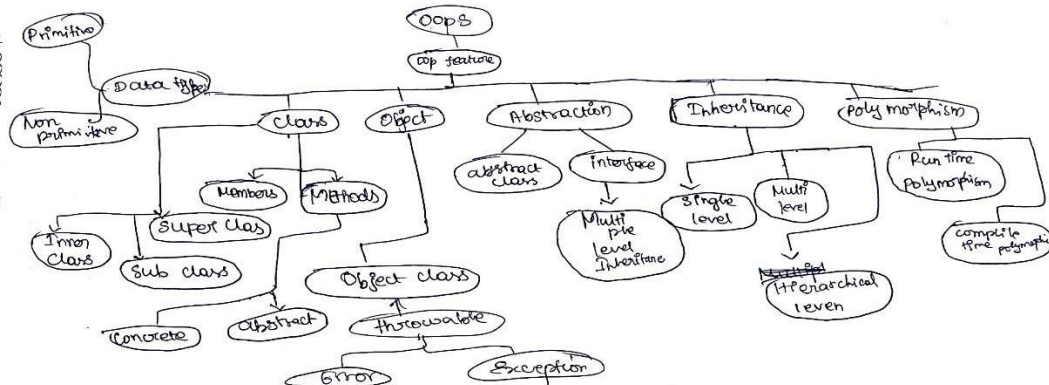
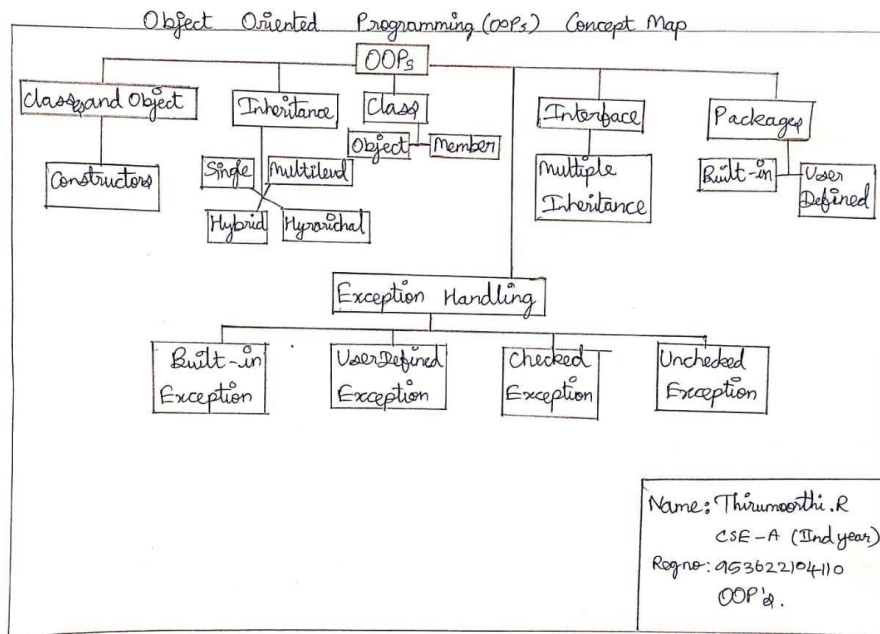
(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO6	PO8
	3	3	1	1	1	1	3
Justification for correlation	The students apply the knowledge of Data Analytics	The students identify the ways to represent the concepts of Data Science					The students apply the knowledge of Data Analytics

Innovative practice	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	2	1	1	3	3	1	1
Justification for correlation	The students identify the ways to represent the concepts of Data Science	The students individually present their ideas	The students communicate the basics Data Analytics concepts in the form of concept map	The Data Science concept are needed in developing the Analysis tool	The Data Science concept are needed in developing the Analysis tool	The concept is used in security applications	The concept is used in solving problems in Data Science

• Images / Screenshot of the practice:



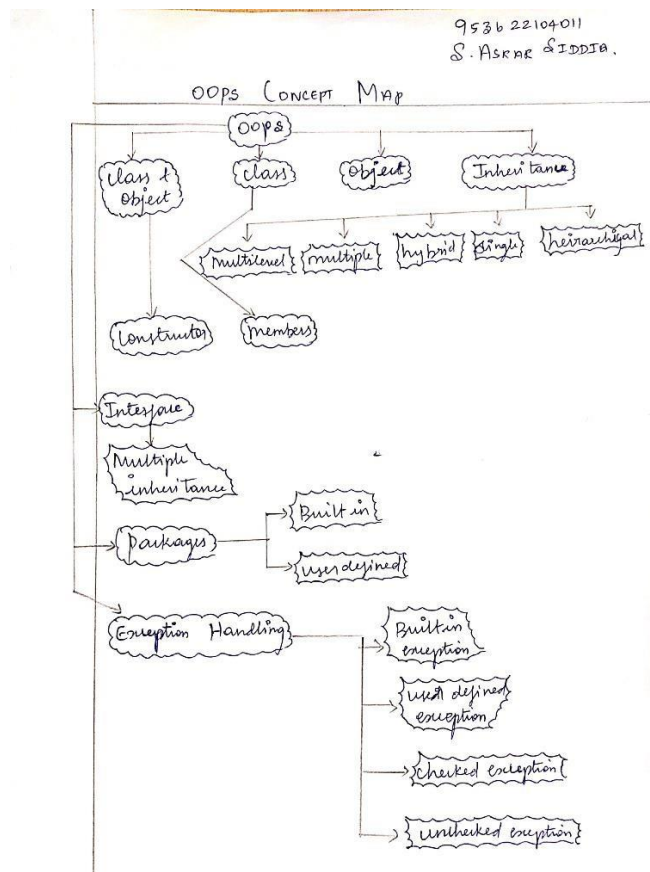


Fig. 1: Sample Concept Map

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- All the students are very much interested in drawing the features of Building Model in Data Analytics as a concept map.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- This activity helps them to summarize the entire concepts in one picture.
- The level of understanding can be analyzed by seeing the concept map they have drawn.

❖ **Challenges faced in implementation:**

- Some of the students need more time to complete the map.
- Some students only represent few concepts in the map.

References:

- <https://www.youtube.com/watch?v=iILLdQuer0E>

Faculty Name / CSBS

HOD / CSBS